# FP&A Commentary Copilot

AI-powered financial performance analysis and reporting

**AI Engineering Capstone**

**Presented by Anudev S D**

Python | OpenAI API | Tool calling | RAG (ChromaDB) | Guardrails

# The problem

- Finance teams spend hours turning P&L numbers into executive-ready commentary every month.
- LLMs write fluent text, but they invent numbers and assert causes nobody has evidence for.
- In finance, one wrong figure or an unsupported claim damages trust in the whole report.

**Goal**
A copilot that drafts the monthly narrative, explains the drivers, and refuses to state anything it cannot trace back to the data or to the finance policy documents.

# What the copilot produces

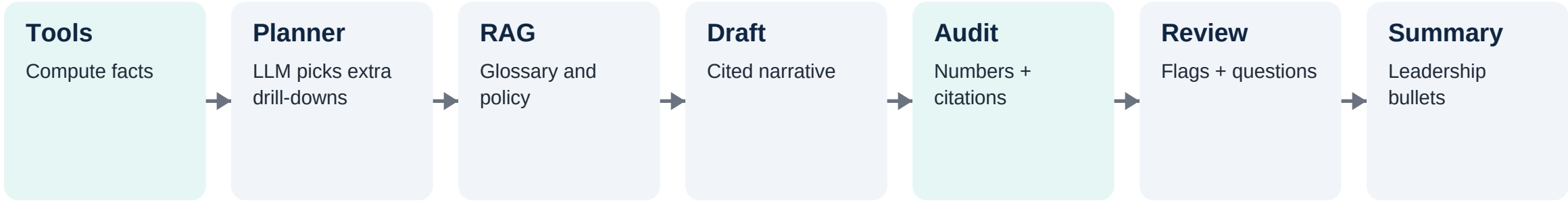
**1. Variance analysis**
Actual vs Budget, Forecast or Prior Year. Variance, variance %, favourable or unfavourable, top drivers by segment, product and country.

**2. Draft narrative**
Structured commentary: overview, revenue and discounts, cost and margin, key drivers, data notes. Every number is cited.

**3. Review Mode**
Flags unsupported claims, missing data and weak assumptions, then asks the analyst follow-up questions.

**4. Leadership Summary**
A 3 to 5 bullet version for the monthly reporting pack, audited the same way as the draft.

# Architecture




- Code guarantees the essential facts. The LLM adds extra drill-downs using OpenAI function calling.
- Every tool result gets an id (F1, F2, ...). The narrative cites these ids, and finance docs as [D:...].
- If the audit finds a bad number, the model gets one repair pass. Anything left shows up as a flag.

# Data

- Source: Financial Sample workbook, 700 transactions from Sep 2013 to Dec 2014.
- 5 segments, 5 countries, 6 products. Accounts: Gross Sales, Discounts, COGS, Units Sold.
- Net Sales = Gross Sales - Discounts. Gross Profit = Net Sales - COGS.
- The file has no Budget or Forecast, so both are simulated with a fixed seed and disclosed in every report.
- Prior Year is Actual shifted by 12 months, so it exists only from Sep 2014 and partial coverage is flagged.

**Why simulate on purpose**

The simulation plants known patterns (some segments over or under plan), giving an independent answer key to test driver correctness.

# Tool calling: sample output (Dec 2014 vs simulated Budget)

| Line | Actual | Budget | Variance | Var % | Label |
|---|---|---|---|---|---|
| **Gross Sales** | $12.51M | $11.39M | $1.12M | 9.8% | favourable |
| **Discounts** | $0.51M | $0.57M | $-0.06M | -10.5% | favourable |
| **Net Sales** | $12.00M | $10.82M | $1.18M | 10.9% | favourable |
| **COGS** | $9.97M | $9.10M | $0.87M | 9.6% | unfavourable |
| **Gross Profit** | $2.03M | $1.72M | $0.31M | 17.8% | favourable |

- Variance is always Actual minus comparison. Costs are favourable when LOWER than the comparison.
- The LLM never calculates. It can only quote these tool results, rounded.
- Tools: get_pnl_summary, get_top_drivers, get_trend, get_data_quality.

# RAG: grounding in finance definitions

- Knowledge base: a finance glossary and accounting notes, split by heading into small chunks.
- Stored in ChromaDB with OpenAI embeddings (text-embedding-3-small).
- Pinned chunks are always included: narrative rules, materiality threshold, sign conventions.
- Other chunks are retrieved by meaning, using queries built from the facts (e.g. Discounts definition).
- Citations like [D:glossary#gross-margin-percent] let a reviewer check every definition used.

**Effect**

Consistent wording and definitions every month, and a built-in rule: never assert a cause the data does not show.

# Guardrails: no fabricated numbers

**Number check**
Every figure in the text must match a tool output after rounding. $12.4M can match 12,378,765.

**Citation check**
Every numeric sentence needs a valid [F#] citation. Unknown ids are flagged.

**Unverified notes**
Analyst notes are labelled unverified and never copied as fact or as numbers.

**Label control**
Favourable and unfavourable come from code, not from the model.

Example: a draft saying 'Net Sales were $99.9M' is caught, repaired once, and flagged if it persists.

# Review Mode and Leadership Summary

**Review Mode**

Combines three sources: data checks (coverage, unbudgeted lines, concentration), guardrail results, and LLM judgement (unsupported claims, missed material variances, inconsistencies). Output is a severity-ranked table plus follow-up questions for the analyst.

**Leadership Summary**

3 to 5 bullets of at most 30 words each: headline vs comparison, biggest drivers, and the key caveat. It is audited with the same number and citation checks, then citations are stripped for the reporting pack.

**All output is marked DRAFT for FP&A review.**

# Evaluation

- Driver correctness (verified): top deltas and totals match an independent recomputation from the raw Excel file.
- 78 automated tests pass: tools, guardrails, retrieval and a full pipeline run with a scripted LLM.
- Narrative metrics via the evaluation script on 6 test cases, including an adversarial case with an unverified cause and invented numbers:

| | | |
|---|---|---|
| **Ungrounded numbers (target 0)** | **Invalid citations (target 0)** | **Driver recall in Key Drivers** |
| **Expected review flags raised** | **LLM judge: groundedness, clarity, consistency** | **Adversarial: note stated as fact (target no)** |

Live-run results: add the table from outputs/eval_report.md here.

# Demo walkthrough

- 1. Choose period, comparison (Budget, Forecast, Prior Year) and scope (month or year to date).
- 2. Generate. The tools compute facts, RAG retrieves policy, the model drafts and the audit runs.
- 3. Read the Leadership Summary, then the cited commentary.
- 4. Open Review Mode to see flags and follow-up questions.
- 5. Open Facts to trace any number back to its tool output.
- 6. Adversarial test: type an unverified competitor claim in the analyst note and watch it get flagged.

# Limitations and next steps

**Limitations**

Only Gross Sales, Discounts, COGS and units exist, so no opex or EBITDA. Price-volume-mix is not computed. Budget, Forecast and policies are simulated. Direction words are not machine-checked, only numbers and labels. Currency is not stated in the source.

**Next steps**

Price-volume-mix decomposition. Operating expense accounts. Multi-month trend narratives. Human edit and approve workflow. A FastAPI service wrapper for integration with reporting tools.

# Thank you

**Anudev S D**

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